CFG for 'batch_norm_compute_' function

entry:

%0 = sext i32 %C to i64, !dbg !21
%1 = shl nuw nsw i64 %0, 2, !dbg !21
%2 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !21, !tbaa !22
%A_red = tail call ptr %2(i32 1, i32 %dev_id, i64 %1, i32 2, i32 32), !dbg
. . . !21

call void @llvm.assume(i1 true) ["align"(ptr %A_red, i64 64)], !dbg !21
%3 = icmp eq ptr %A_red, null, !dbg !21
br i1 %3, label %common.ret, label %if_end, !dbg !21, !prof !26

T F

common.ret:

%common.ret.op = phi i32 [-1, %entry], [-1, %if_end], [-1, %if_end2
. . .], [-1, %if_end7], [-1, %for_end_ax0140], [-1, %if_end155], [-1,
. . . %if_end158], [%., %if_end161]
ret i32 %common.ret.op, !dbg !21

if_end:

%4 = mul i32 %H, %N, !dbg !21
%5 = mul i32 %4, %C, !dbg !21
%6 = mul nsw i32 %5, %W, !dbg !21
%7 = sext i32 %6 to i64, !dbg !21
%8 = shl nuw nsw i64 %7, 2, !dbg !21
%9 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !21, !tbaa !22
%T_subtract = tail call ptr %9(i32 1, i32 %dev_id, i64 %8, i32 2, i32 32),
. . . !dbg !21
call void @llvm.assume(i1 true) ["align"(ptr %T_subtract, i64 64)], !dbg
. . . !21

%10 = icmp eq ptr %T_subtract, null, !dbg !21
br i1 %10, label %common.ret, label %if_end2, !dbg !21, !prof !26

T F

if_end2:

%11 = tail call i32 @llvm.smax.i32(i32 %6, i32 %C), !dbg !21
%12 = sext i32 %11 to i64, !dbg !21
%13 = shl nuw nsw i64 %12, 2, !dbg !21
%14 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !21, !tbaa !22
%T_subtract5 = tail call ptr %14(i32 1, i32 %dev_id, i64 %13, i32 2, i32
. . . 32), !dbg !21
call void @llvm.assume(i1 true) ["align"(ptr %T_subtract5, i64 64)], !dbg
. . . !21

%15 = icmp eq ptr %T_subtract5, null, !dbg !21
br i1 %15, label %common.ret, label %if_end7, !dbg !21, !prof !26

T F

if_end7:

%16 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !21, !tbaa
. . . !22

%T_subtract10 = tail call ptr %16(i32 1, i32 %dev_id, i64 %13, i32 2, i32
. . . 32), !dbg !21
call void @llvm.assume(i1 true) ["align"(ptr %T_subtract10, i64 64)], !dbg
. . . !21

%17 = icmp eq ptr %T_subtract10, null, !dbg !21
br i1 %17, label %common.ret, label %for_begin_ax0.preheader, !dbg !21,
. . . !prof !26

T F

for_begin_ax0.preheader:

%18 = icmp sgt i32 %C, 0, !dbg !21
br i1 %18, label %for_begin_k0.preheader.lr.ph, label
. . . %for_begin_ax020.preheader, !dbg !21, !prof !29

T F

for_begin_k0.preheader.lr.ph:

%19 = icmp slt i32 %N, 1
%20 = icmp slt i32 %W, 1
%21 = icmp slt i32 %H, 1
%or.cond.not577 = select i1 %19, i1 true, i1 %21
%brmerge = select i1 %or.cond.not577, i1 true, i1 %20
br i1 %brmerge, label %for_body_ax017.lr.ph, label
. . . %for_begin_k0.preheader.us.us.us.preheader, !prof !30

T F

for_begin_ax020.preheader:

%51 = icmp slt i32 %N, 1, !dbg !21
br i1 %51, label %for_begin_ax072.preheader, label
. . . %for_begin_ax1.preheader.lr.ph, !dbg !21, !prof !36

T F

for_body_ax017.lr.ph:

%factor.op.mul = mul i32 %H, %W, !dbg !21
%.reass = mul i32 %factor.op.mul, %N
%47 = sitofp i32 %.reass to float
%wide.trip.count355 = zext nneg i32 %C to i64, !dbg !21
%min.iters.check = icmp ult i32 %C, 4, !dbg !21
br i1 %min.iters.check, label %for_body_ax017.preheader, label %vector.ph,
. . . !dbg !21, !prof !36

T F

for_begin_k0.preheader.us.us.us.preheader:

%22 = zext nneg i32 %W to i64, !dbg !21
%23 = zext nneg i32 %H to i64, !dbg !21
%wide.trip.count350 = zext nneg i32 %C to i64, !dbg !21
%wide.trip.count345 = zext nneg i32 %N to i64
%wide.trip.count340 = zext nneg i32 %H to i64
%wide.trip.count = zext nneg i32 %W to i64
%xtraiter = and i64 %wide.trip.count, 1
%24 = icmp eq i32 %W, 1
%unroll_iter = and i64 %wide.trip.count, 2147483646
%lcmp.mod.not = icmp eq i64 %xtraiter, 0
br label %for_begin_k0.preheader.us.us.us, !dbg !21

for_begin_ax072.preheader:

br i1 %18, label %for_begin_k076.preheader.lr.ph, label
. . . %for_begin_ax098.preheader, !dbg !21, !prof !69

T F

for_begin_ax1.preheader.lr.ph:

%52 = icmp sgt i32 %W, 0
br i1 %18, label %for_begin_ax1.preheader.lr.ph.split.us, label
. . . %for_begin_ax1102.preheader.lr.ph, !prof !29

T F

for_body_ax017.preheader:

%indvars.iv352.ph = phi i64 [0, %for_body_ax017.lr.ph], [%n.vec,
. . . %middle.block]
br label %for_body_ax017, !dbg !21

vector.ph:

%n.vec = and i64 %wide.trip.count355, 2147483644, !dbg !21
%broadcast.splatinsert = insertelement <4 x float> poison, float %47, i64 0,
. . . !dbg !21
%broadcast.splat = shufflevector <4 x float> %broadcast.splatinsert, <4 x
. . . float> poison, <4 x i32> zeroinitializer, !dbg !21
br label %vector.body, !dbg !21

for_begin_k0.preheader.us.us.us:

%indvars.iv347 = phi i64 [0, %for_begin_k0.preheader.us.us.us.preheader
. . .], [%indvars.iv.next348,
. . . %for_begin_k0.for_end_k0_crit_edge.split.us.split.us.us.us.us]
%25 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv347
%.promoted.us.us.us = load float, ptr %25, align 4, !tbaa !32
br label %for_begin_k2.preheader.us.us.us.us.us, !dbg !21

for_begin_k2.preheader.us.us.us.us.us:

%indvars.iv342 = phi i64 [%indvars.iv.next343,
. . . %for_begin_k2.for_end_k2_crit_edge.split.us.us.us.us.us.us], [0,
. . . %for_begin_k0.preheader.us.us.us]
%.lcssa.us.lcssa.us.us178.us.us.us = phi float [%.lcssa727,
. . . %for_begin_k2.for_end_k2_crit_edge.split.us.us.us.us.us.us], [
. . . %.promoted.us.us.us, %for_begin_k0.preheader.us.us.us]
%26 = mul nuw nsw i64 %indvars.iv342, %0
%27 = add nuw nsw i64 %26, %indvars.iv347
%28 = mul nuw nsw i64 %27, %23
br label %for_begin_k3.preheader.us.us.us.us.us.us, !dbg !21

for_begin_k3.preheader.us.us.us.us.us.us:

%indvars.iv337 = phi i64 [%indvars.iv.next338,
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us], [0,
. . . %for_begin_k2.preheader.us.us.us.us.us]
%.lcssa.us171.us.us.us.us.us = phi float [%.lcssa727,
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us], [
. . . %.lcssa.us.lcssa.us.us178.us.us.us, %for_begin_k2.preheader.us.us.us.us.us]
%29 = or i64 %indvars.iv337, %indvars.iv342
%30 = add nuw nsw i64 %indvars.iv337, %28
%31 = mul nuw nsw i64 %30, %22
%invariant.gep = getelementptr float, ptr %A, i64 %31, !dbg !21
br i1 %24, label
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us.unr-lcssa, label
. . . %for_body_k3.us.us.us.us.us.us, !dbg !21, !prof !36

T F

for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us.unr-lcssa:

%.lcssa727.ph = phi float [poison,
. . . %for_begin_k3.preheader.us.us.us.us.us.us], [%40,
. . . %for_body_k3.us.us.us.us.us.us]
%indvars.iv.unr = phi i64 [0, %for_begin_k3.preheader.us.us.us.us.us.us],
. . . [%indvars.iv.next.1, %for_body_k3.us.us.us.us.us.us]
%.unr = phi float [%.lcssa.us171.us.us.us.us.us,
. . . %for_begin_k3.preheader.us.us.us.us.us.us], [%40,
. . . %for_body_k3.us.us.us.us.us.us]
br i1 %lcmp.mod.not, label
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us, label
. . . %for_body_k3.us.us.us.us.us.us.epil, !dbg !21, !prof !40

T F

for_body_k3.us.us.us.us.us.us:

%indvars.iv = phi i64 [%indvars.iv.next.1, %for_body_k3.us.us.us.us.us.us
. . .], [0, %for_begin_k3.preheader.us.us.us.us.us.us]
%32 = phi float [%40, %for_body_k3.us.us.us.us.us.us], [
. . . %.lcssa.us171.us.us.us.us.us, %for_begin_k3.preheader.us.us.us.us.us.us]
%niter = phi i64 [%niter.next.1, %for_body_k3.us.us.us.us.us.us], [0,
. . . %for_begin_k3.preheader.us.us.us.us.us.us]
%33 = or i64 %29, %indvars.iv, !dbg !21
%34 = and i64 %33, 4294967295, !dbg !21
%35 = icmp eq i64 %34, 0, !dbg !21
%36 = select i1 %35, float 0.000000e+00, float %32, !dbg !21, !prof !26
%gep = getelementptr float, ptr %invariant.gep, i64 %indvars.iv, !dbg !21
%37 = load float, ptr %gep, align 4, !dbg !21, !tbaa !37
%38 = fadd float %37, %36, !dbg !21
%indvars.iv.next = or disjoint i64 %indvars.iv, 1, !dbg !21
%gep.1 = getelementptr float, ptr %invariant.gep, i64 %indvars.iv.next, !dbg
. . . !21

%39 = load float, ptr %gep.1, align 4, !dbg !21, !tbaa !37
%40 = fadd float %39, %38, !dbg !21
%indvars.iv.next.1 = add nuw nsw i64 %indvars.iv, 2, !dbg !21
%niter.next.1 = add i64 %niter, 2, !dbg !21
%niter.ncmp.1 = icmp eq i64 %niter.next.1, %unroll_iter, !dbg !21
br i1 %niter.ncmp.1, label
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us.unr-lcssa, label
. . . %for_body_k3.us.us.us.us.us.us, !dbg !21, !prof !39

T F

for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us:

%.lcssa727 = phi float [%.lcssa727.ph,
. . . %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us.unr-lcssa], [%46,
. . . %for_body_k3.us.us.us.us.us.us.epil], !dbg !21
%indvars.iv.next338 = add nuw nsw i64 %indvars.iv337, 1, !dbg !21
%exitcond341.not = icmp eq i64 %indvars.iv.next338, %wide.trip.count340,
. . . !dbg !21
br i1 %exitcond341.not, label
. . . %for_begin_k2.for_end_k2_crit_edge.split.us.us.us.us.us.us, label
. . . %for_begin_k3.preheader.us.us.us.us.us.us, !dbg !21, !prof !41

T F

for_body_k3.us.us.us.us.us.us.epil:

%41 = or i64 %29, %indvars.iv.unr, !dbg !21
%42 = and i64 %41, 4294967295, !dbg !21
%43 = icmp eq i64 %42, 0, !dbg !21
%44 = select i1 %43, float 0.000000e+00, float %.unr, !dbg !21, !prof !26
%gep.epil = getelementptr float, ptr %invariant.gep, i64 %indvars.iv.unr,
. . . !dbg !21
%45 = load float, ptr %gep.epil, align 4, !dbg !21, !tbaa !37
%46 = fadd float %45, %44, !dbg !21
br label %for_begin_k3.for_end_k3_crit_edge.us.us.us.us.us.us, !dbg !21

for_begin_k2.for_end_k2_crit_edge.split.us.us.us.us.us.us:

%indvars.iv.next343 = add nuw nsw i64 %indvars.iv342, 1, !dbg !21
%exitcond346.not = icmp eq i64 %indvars.iv.next343, %wide.trip.count345,
. . . !dbg !21
br i1 %exitcond346.not, label
. . . %for_begin_k0.for_end_k0_crit_edge.split.us.split.us.us.us.us, label
. . . %for_begin_k2.preheader.us.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_k0.for_end_k0_crit_edge.split.us.split.us.us.us.us:

store float %.lcssa727, ptr %25, align 4, !tbaa !32
%indvars.iv.next348 = add nuw nsw i64 %indvars.iv347, 1, !dbg !21
%exitcond351.not = icmp eq i64 %indvars.iv.next348, %wide.trip.count350,
. . . !dbg !21
br i1 %exitcond351.not, label %for_body_ax017.lr.ph, label
. . . %for_begin_k0.preheader.us.us.us, !dbg !21, !prof !41

T F

for_body_ax017:

%indvars.iv352 = phi i64 [%indvars.iv.next353, %for_body_ax017], [
. . . %indvars.iv352.ph, %for_body_ax017.preheader]
%76 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv352, !dbg
. . . !21

%77 = load float, ptr %76, align 4, !dbg !21, !tbaa !32
%78 = fdiv float %77, %47, !dbg !21
store float %78, ptr %76, align 4, !dbg !21, !tbaa !32
%indvars.iv.next353 = add nuw nsw i64 %indvars.iv352, 1, !dbg !21
%exitcond356.not = icmp eq i64 %indvars.iv.next353, %wide.trip.count355,
. . . !dbg !21
br i1 %exitcond356.not, label %for_begin_ax020.preheader, label
. . . %for_body_ax017, !dbg !21, !prof !55, !llvm.loop !57

T F

vector.body:

%index = phi i64 [0, %vector.ph], [%index.next, %vector.body], !dbg !21
%48 = getelementptr inbounds nuw float, ptr %A_red, i64 %index, !dbg !21
%wide.load = load <4 x float>, ptr %48, align 16, !dbg !21, !tbaa !32
%49 = fdiv <4 x float> %wide.load, %broadcast.splat, !dbg !21
store <4 x float> %49, ptr %48, align 16, !dbg !21, !tbaa !32
%index.next = add nuw i64 %index, 4, !dbg !21
%50 = icmp eq i64 %index.next, %n.vec, !dbg !21
br i1 %50, label %middle.block, label %vector.body, !dbg !21, !prof !42,
. . . !llvm.loop !43

T F

middle.block:

%cmp.n = icmp eq i64 %n.vec, %wide.trip.count355, !dbg !21
br i1 %cmp.n, label %for_begin_ax020.preheader, label
. . . %for_body_ax017.preheader, !dbg !21, !prof !46

T F

for_begin_k076.preheader.lr.ph:

%149 = icmp slt i32 %W, 1
%150 = icmp slt i32 %H, 1
%or.cond563.not580 = select i1 %51, i1 true, i1 %150
%brmerge565 = select i1 %or.cond563.not580, i1 true, i1 %149
br i1 %brmerge565, label %for_body_ax091.lr.ph, label
. . . %for_begin_k076.preheader.us.us.us.preheader, !prof !30

T F

for_begin_ax098.preheader:

br i1 %51, label %for_begin_ax1114.preheader, label
. . . %for_begin_ax1102.preheader.lr.ph, !dbg !21, !prof !69

T F

for_begin_ax1.preheader.lr.ph.split.us:

%53 = icmp slt i32 %H, 1
br i1 %53, label %for_begin_ax128.preheader.us.preheader, label
. . . %for_begin_ax1.preheader.us.us.preheader, !prof !36

T F

for_begin_ax1102.preheader.lr.ph:

%194 = icmp slt i32 %H, 1
%195 = icmp slt i32 %W, 1
%196 = sext i32 %W to i64, !dbg !21
%197 = sext i32 %H to i64, !dbg !21
%wide.trip.count490 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count485 = zext nneg i32 %C to i64
%wide.trip.count480 = zext nneg i32 %H to i64
%wide.trip.count475 = zext i32 %W to i64
%brmerge568 = select i1 %194, i1 true, i1 %195
%min.iters.check683 = icmp ult i32 %W, 4
%n.vec686 = and i64 %wide.trip.count475, 2147483644
%cmp.n693 = icmp eq i64 %n.vec686, %wide.trip.count475
br label %for_begin_ax1102.preheader, !dbg !21

for_begin_ax128.preheader.us.preheader:

%79 = icmp slt i32 %W, 1
%80 = sext i32 %W to i64
%81 = sext i32 %H to i64
%wide.trip.count395 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count390 = zext nneg i32 %C to i64
%wide.trip.count385 = zext nneg i32 %H to i64
%wide.trip.count380 = zext i32 %W to i64
%brmerge556 = select i1 %53, i1 true, i1 %79
%min.iters.check599 = icmp ult i32 %W, 8
%n.vec602 = and i64 %wide.trip.count380, 2147483640
%cmp.n610 = icmp eq i64 %n.vec602, %wide.trip.count380
br label %for_begin_ax128.preheader.us

for_begin_ax1.preheader.us.us.preheader:

%54 = sext i32 %W to i64
%55 = zext nneg i32 %H to i64
%wide.trip.count375 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count370 = zext nneg i32 %C to i64
%wide.trip.count365 = zext nneg i32 %H to i64
%wide.trip.count360 = zext i32 %W to i64
%min.iters.check584 = icmp ult i32 %W, 8
%n.vec587 = and i64 %wide.trip.count360, 2147483640
%cmp.n595 = icmp eq i64 %n.vec587, %wide.trip.count360
br label %for_begin_ax1.preheader.us.us

for_begin_ax1102.preheader:

%indvars.iv487 = phi i64 [0, %for_begin_ax1102.preheader.lr.ph], [
. . . %indvars.iv.next488, %for_end_ax1104]
br i1 %18, label %for_begin_ax2106.preheader.lr.ph, label %for_end_ax1104,
. . . !dbg !21, !prof !29

T F

for_begin_ax128.preheader.us:

%indvars.iv392 = phi i64 [0, %for_begin_ax128.preheader.us.preheader], [
. . . %indvars.iv.next393, %for_begin_ax128.for_end_ax130_crit_edge.us]
%82 = mul nuw nsw i64 %indvars.iv392, %0
br i1 %brmerge556, label %for_begin_ax128.for_end_ax130_crit_edge.us, label
. . . %for_begin_ax232.preheader.us.us.us, !prof !58

T F

for_begin_ax1.preheader.us.us:

%indvars.iv372 = phi i64 [0, %for_begin_ax1.preheader.us.us.preheader],
. . . [%indvars.iv.next373, %for_begin_ax1.for_end_ax1_crit_edge.split.us.us.us]
%56 = mul nuw nsw i64 %indvars.iv372, %0
br i1 %52, label %for_begin_ax2.preheader.us.us.us.us, label
. . . %for_begin_ax1.for_end_ax1_crit_edge.split.us.us.us, !prof !29

T F

for_begin_ax2.preheader.us.us.us.us:

%indvars.iv367 = phi i64 [%indvars.iv.next368,
. . . %for_begin_ax2.for_end_ax2_crit_edge.split.us.us.us.us.us], [0,
. . . %for_begin_ax1.preheader.us.us]
%57 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv367
%58 = add nuw nsw i64 %indvars.iv367, %56
%59 = mul nuw nsw i64 %58, %55
%60 = load float, ptr %57, align 4, !tbaa !32
%broadcast.splatinsert592 = insertelement <4 x float> poison, float %60, i64
. . . 0

%broadcast.splat593 = shufflevector <4 x float> %broadcast.splatinsert592,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax3.preheader.us.us.us.us.us, !dbg !21

for_begin_ax1.for_end_ax1_crit_edge.split.us.us.us:

%indvars.iv.next373 = add nuw nsw i64 %indvars.iv372, 1, !dbg !21
%exitcond376.not = icmp eq i64 %indvars.iv.next373, %wide.trip.count375,
. . . !dbg !21
br i1 %exitcond376.not, label %for_begin_ax128.preheader.us.preheader, label
. . . %for_begin_ax1.preheader.us.us, !dbg !21, !prof !41

T F

for_begin_ax3.preheader.us.us.us.us.us:

%indvars.iv362 = phi i64 [%indvars.iv.next363,
. . . %for_begin_ax3.for_end_ax3_crit_edge.us.us.us.us.us], [0,
. . . %for_begin_ax2.preheader.us.us.us.us]
%61 = add nuw nsw i64 %indvars.iv362, %59
%62 = mul nuw nsw i64 %61, %54
br i1 %min.iters.check584, label %for_body_ax3.us.us.us.us.us.preheader,
. . . label %vector.body588, !dbg !21, !prof !36

T F

for_body_ax3.us.us.us.us.us.preheader:

%indvars.iv357.ph = phi i64 [0, %for_begin_ax3.preheader.us.us.us.us.us
. . .], [%n.vec587, %middle.block582]
br label %for_body_ax3.us.us.us.us.us, !dbg !21

vector.body588:

%index589 = phi i64 [%index.next594, %vector.body588], [0,
. . . %for_begin_ax3.preheader.us.us.us.us.us], !dbg !21
%63 = add nuw nsw i64 %index589, %62, !dbg !21
%64 = getelementptr inbounds float, ptr %A, i64 %63, !dbg !21
%65 = getelementptr inbounds nuw i8, ptr %64, i64 16, !dbg !21
%wide.load590 = load <4 x float>, ptr %64, align 4, !dbg !21, !tbaa !37
%wide.load591 = load <4 x float>, ptr %65, align 4, !dbg !21, !tbaa !37
%66 = fsub <4 x float> %wide.load590, %broadcast.splat593, !dbg !21
%67 = fsub <4 x float> %wide.load591, %broadcast.splat593, !dbg !21
%68 = getelementptr inbounds float, ptr %T_subtract, i64 %63, !dbg !21
%69 = getelementptr inbounds nuw i8, ptr %68, i64 16, !dbg !21
store <4 x float> %66, ptr %68, align 4, !dbg !21, !tbaa !50
store <4 x float> %67, ptr %69, align 4, !dbg !21, !tbaa !50
%index.next594 = add nuw i64 %index589, 8, !dbg !21
%70 = icmp eq i64 %index.next594, %n.vec587, !dbg !21
br i1 %70, label %middle.block582, label %vector.body588, !dbg !21, !prof
. . . !52, !llvm.loop !53

T F

for_body_ax3.us.us.us.us.us:

%indvars.iv357 = phi i64 [%indvars.iv.next358,
. . . %for_body_ax3.us.us.us.us.us], [%indvars.iv357.ph,
. . . %for_body_ax3.us.us.us.us.us.preheader]
%71 = add nuw nsw i64 %indvars.iv357, %62, !dbg !21
%72 = getelementptr inbounds float, ptr %A, i64 %71, !dbg !21
%73 = load float, ptr %72, align 4, !dbg !21, !tbaa !37
%74 = fsub float %73, %60, !dbg !21
%75 = getelementptr inbounds float, ptr %T_subtract, i64 %71, !dbg !21
store float %74, ptr %75, align 4, !dbg !21, !tbaa !50
%indvars.iv.next358 = add nuw nsw i64 %indvars.iv357, 1, !dbg !21
%exitcond361.not = icmp eq i64 %indvars.iv.next358, %wide.trip.count360,
. . . !dbg !21
br i1 %exitcond361.not, label
. . . %for_begin_ax3.for_end_ax3_crit_edge.us.us.us.us.us, label
. . . %for_body_ax3.us.us.us.us.us, !dbg !21, !prof !55, !llvm.loop !56

T F

middle.block582:

br i1 %cmp.n595, label
. . . %for_begin_ax3.for_end_ax3_crit_edge.us.us.us.us.us, label
. . . %for_body_ax3.us.us.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax3.for_end_ax3_crit_edge.us.us.us.us.us:

%indvars.iv.next363 = add nuw nsw i64 %indvars.iv362, 1, !dbg !21
%exitcond366.not = icmp eq i64 %indvars.iv.next363, %wide.trip.count365,
. . . !dbg !21
br i1 %exitcond366.not, label
. . . %for_begin_ax2.for_end_ax2_crit_edge.split.us.us.us.us.us, label
. . . %for_begin_ax3.preheader.us.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax2.for_end_ax2_crit_edge.split.us.us.us.us.us:

%indvars.iv.next368 = add nuw nsw i64 %indvars.iv367, 1, !dbg !21
%exitcond371.not = icmp eq i64 %indvars.iv.next368, %wide.trip.count370,
. . . !dbg !21
br i1 %exitcond371.not, label
. . . %for_begin_ax1.for_end_ax1_crit_edge.split.us.us.us, label
. . . %for_begin_ax2.preheader.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax128.for_end_ax130_crit_edge.us:

%indvars.iv.next393 = add nuw nsw i64 %indvars.iv392, 1, !dbg !21
%exitcond396.not = icmp eq i64 %indvars.iv.next393, %wide.trip.count395,
. . . !dbg !21
br i1 %exitcond396.not, label %for_begin_ax144.preheader.lr.ph, label
. . . %for_begin_ax128.preheader.us, !dbg !21, !prof !41

T F

for_begin_ax232.preheader.us.us.us:

%indvars.iv387 = phi i64 [%indvars.iv.next388,
. . . %for_begin_ax232.for_end_ax234_crit_edge.split.us.us.us.us], [0,
. . . %for_begin_ax128.preheader.us]
%83 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv387
%84 = add nuw nsw i64 %indvars.iv387, %82
%85 = mul nuw nsw i64 %84, %81
%86 = load float, ptr %83, align 4, !tbaa !32
%broadcast.splatinsert607 = insertelement <4 x float> poison, float %86, i64
. . . 0

%broadcast.splat608 = shufflevector <4 x float> %broadcast.splatinsert607,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax336.preheader.us.us.us.us, !dbg !21

for_begin_ax144.preheader.lr.ph:

%102 = icmp slt i32 %H, 1
%103 = icmp slt i32 %W, 1
%104 = sext i32 %W to i64
%105 = sext i32 %H to i64
%wide.trip.count415 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count410 = zext nneg i32 %C to i64
%wide.trip.count405 = zext nneg i32 %H to i64
%wide.trip.count400 = zext nneg i32 %W to i64
%brmerge559 = select i1 %102, i1 true, i1 %103
%min.iters.check614 = icmp ult i32 %W, 8
%n.vec617 = and i64 %wide.trip.count380, 2147483640
%cmp.n625 = icmp eq i64 %n.vec617, %wide.trip.count380
br label %for_begin_ax144.preheader.us

for_begin_ax336.preheader.us.us.us.us:

%indvars.iv382 = phi i64 [%indvars.iv.next383,
. . . %for_begin_ax336.for_end_ax338_crit_edge.us.us.us.us], [0,
. . . %for_begin_ax232.preheader.us.us.us]
%87 = add nuw nsw i64 %indvars.iv382, %85
%88 = mul nuw nsw i64 %87, %80
br i1 %min.iters.check599, label %for_body_ax337.us.us.us.us.preheader,
. . . label %vector.body603, !dbg !21, !prof !36

T F

for_begin_ax144.preheader.us:

%indvars.iv412 = phi i64 [0, %for_begin_ax144.preheader.lr.ph], [
. . . %indvars.iv.next413, %for_begin_ax144.for_end_ax146_crit_edge.us]
%106 = mul nuw nsw i64 %indvars.iv412, %0
br i1 %brmerge559, label %for_begin_ax144.for_end_ax146_crit_edge.us, label
. . . %for_begin_ax248.preheader.us.us.us, !prof !58

T F

for_body_ax337.us.us.us.us.preheader:

%indvars.iv377.ph = phi i64 [0, %for_begin_ax336.preheader.us.us.us.us],
. . . [%n.vec602, %middle.block597]
br label %for_body_ax337.us.us.us.us, !dbg !21

vector.body603:

%index604 = phi i64 [%index.next609, %vector.body603], [0,
. . . %for_begin_ax336.preheader.us.us.us.us], !dbg !21
%89 = add nuw nsw i64 %index604, %88, !dbg !21
%90 = getelementptr inbounds float, ptr %A, i64 %89, !dbg !21
%91 = getelementptr inbounds nuw i8, ptr %90, i64 16, !dbg !21
%wide.load605 = load <4 x float>, ptr %90, align 4, !dbg !21, !tbaa !37
%wide.load606 = load <4 x float>, ptr %91, align 4, !dbg !21, !tbaa !37
%92 = fsub <4 x float> %wide.load605, %broadcast.splat608, !dbg !21
%93 = fsub <4 x float> %wide.load606, %broadcast.splat608, !dbg !21
%94 = getelementptr inbounds float, ptr %T_subtract5, i64 %89, !dbg !21
%95 = getelementptr inbounds nuw i8, ptr %94, i64 16, !dbg !21
store <4 x float> %92, ptr %94, align 4, !dbg !21, !tbaa !59
store <4 x float> %93, ptr %95, align 4, !dbg !21, !tbaa !59
%index.next609 = add nuw i64 %index604, 8, !dbg !21
%96 = icmp eq i64 %index.next609, %n.vec602, !dbg !21
br i1 %96, label %middle.block597, label %vector.body603, !dbg !21, !prof
. . . !52, !llvm.loop !61

T F

for_body_ax337.us.us.us.us:

%indvars.iv377 = phi i64 [%indvars.iv.next378,
. . . %for_body_ax337.us.us.us.us], [%indvars.iv377.ph,
. . . %for_body_ax337.us.us.us.us.preheader]
%97 = add nuw nsw i64 %indvars.iv377, %88, !dbg !21
%98 = getelementptr inbounds float, ptr %A, i64 %97, !dbg !21
%99 = load float, ptr %98, align 4, !dbg !21, !tbaa !37
%100 = fsub float %99, %86, !dbg !21
%101 = getelementptr inbounds float, ptr %T_subtract5, i64 %97, !dbg !21
store float %100, ptr %101, align 4, !dbg !21, !tbaa !59
%indvars.iv.next378 = add nuw nsw i64 %indvars.iv377, 1, !dbg !21
%exitcond381.not = icmp eq i64 %indvars.iv.next378, %wide.trip.count380,
. . . !dbg !21
br i1 %exitcond381.not, label
. . . %for_begin_ax336.for_end_ax338_crit_edge.us.us.us.us, label
. . . %for_body_ax337.us.us.us.us, !dbg !21, !prof !55, !llvm.loop !62

T F

middle.block597:

br i1 %cmp.n610, label
. . . %for_begin_ax336.for_end_ax338_crit_edge.us.us.us.us, label
. . . %for_body_ax337.us.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax336.for_end_ax338_crit_edge.us.us.us.us:

%indvars.iv.next383 = add nuw nsw i64 %indvars.iv382, 1, !dbg !21
%exitcond386.not = icmp eq i64 %indvars.iv.next383, %wide.trip.count385,
. . . !dbg !21
br i1 %exitcond386.not, label
. . . %for_begin_ax232.for_end_ax234_crit_edge.split.us.us.us.us, label
. . . %for_begin_ax336.preheader.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax232.for_end_ax234_crit_edge.split.us.us.us.us:

%indvars.iv.next388 = add nuw nsw i64 %indvars.iv387, 1, !dbg !21
%exitcond391.not = icmp eq i64 %indvars.iv.next388, %wide.trip.count390,
. . . !dbg !21
br i1 %exitcond391.not, label %for_begin_ax128.for_end_ax130_crit_edge.us,
. . . label %for_begin_ax232.preheader.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax144.for_end_ax146_crit_edge.us:

%indvars.iv.next413 = add nuw nsw i64 %indvars.iv412, 1, !dbg !21
%exitcond416.not = icmp eq i64 %indvars.iv.next413, %wide.trip.count415,
. . . !dbg !21
br i1 %exitcond416.not, label %for_begin_ax160.preheader.lr.ph, label
. . . %for_begin_ax144.preheader.us, !dbg !21, !prof !41

T F

for_begin_ax248.preheader.us.us.us:

%indvars.iv407 = phi i64 [%indvars.iv.next408,
. . . %for_begin_ax248.for_end_ax250_crit_edge.split.us.us.us.us], [0,
. . . %for_begin_ax144.preheader.us]
%107 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv407
%108 = add nuw nsw i64 %indvars.iv407, %106
%109 = mul nuw nsw i64 %108, %105
%110 = load float, ptr %107, align 4, !tbaa !32
%broadcast.splatinsert622 = insertelement <4 x float> poison, float %110,
. . . i64 0
%broadcast.splat623 = shufflevector <4 x float> %broadcast.splatinsert622,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax352.preheader.us.us.us.us, !dbg !21

for_begin_ax160.preheader.lr.ph:

%126 = icmp slt i32 %H, 1
%127 = icmp slt i32 %W, 1
%128 = sext i32 %W to i64
%129 = sext i32 %H to i64
%wide.trip.count435 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count430 = zext nneg i32 %C to i64
%wide.trip.count425 = zext nneg i32 %H to i64
%wide.trip.count420 = zext nneg i32 %W to i64
%brmerge562 = select i1 %126, i1 true, i1 %127
%min.iters.check629 = icmp ult i32 %W, 8
%n.vec632 = and i64 %wide.trip.count380, 2147483640
%cmp.n640 = icmp eq i64 %n.vec632, %wide.trip.count380
br label %for_begin_ax160.preheader.us

for_begin_ax352.preheader.us.us.us.us:

%indvars.iv402 = phi i64 [%indvars.iv.next403,
. . . %for_begin_ax352.for_end_ax354_crit_edge.us.us.us.us], [0,
. . . %for_begin_ax248.preheader.us.us.us]
%111 = add nuw nsw i64 %indvars.iv402, %109
%112 = mul nuw nsw i64 %111, %104
br i1 %min.iters.check614, label %for_body_ax353.us.us.us.us.preheader,
. . . label %vector.body618, !dbg !21, !prof !36

T F

for_begin_ax160.preheader.us:

%indvars.iv432 = phi i64 [0, %for_begin_ax160.preheader.lr.ph], [
. . . %indvars.iv.next433, %for_begin_ax160.for_end_ax162_crit_edge.us]
%130 = mul nuw nsw i64 %indvars.iv432, %0
br i1 %brmerge562, label %for_begin_ax160.for_end_ax162_crit_edge.us, label
. . . %for_begin_ax264.preheader.us.us.us, !prof !58

T F

for_body_ax353.us.us.us.us.preheader:

%indvars.iv397.ph = phi i64 [0, %for_begin_ax352.preheader.us.us.us.us],
. . . [%n.vec617, %middle.block612]
br label %for_body_ax353.us.us.us.us, !dbg !21

vector.body618:

%index619 = phi i64 [%index.next624, %vector.body618], [0,
. . . %for_begin_ax352.preheader.us.us.us.us], !dbg !21
%113 = add nuw nsw i64 %index619, %112, !dbg !21
%114 = getelementptr inbounds float, ptr %A, i64 %113, !dbg !21
%115 = getelementptr inbounds nuw i8, ptr %114, i64 16, !dbg !21
%wide.load620 = load <4 x float>, ptr %114, align 4, !dbg !21, !tbaa !37
%wide.load621 = load <4 x float>, ptr %115, align 4, !dbg !21, !tbaa !37
%116 = fsub <4 x float> %wide.load620, %broadcast.splat623, !dbg !21
%117 = fsub <4 x float> %wide.load621, %broadcast.splat623, !dbg !21
%118 = getelementptr inbounds float, ptr %T_subtract10, i64 %113, !dbg !21
%119 = getelementptr inbounds nuw i8, ptr %118, i64 16, !dbg !21
store <4 x float> %116, ptr %118, align 4, !dbg !21, !tbaa !63
store <4 x float> %117, ptr %119, align 4, !dbg !21, !tbaa !63
%index.next624 = add nuw i64 %index619, 8, !dbg !21
%120 = icmp eq i64 %index.next624, %n.vec617, !dbg !21
br i1 %120, label %middle.block612, label %vector.body618, !dbg !21, !prof
. . . !52, !llvm.loop !65

T F

for_body_ax353.us.us.us.us:

%indvars.iv397 = phi i64 [%indvars.iv.next398,
. . . %for_body_ax353.us.us.us.us], [%indvars.iv397.ph,
. . . %for_body_ax353.us.us.us.us.preheader]
%121 = add nuw nsw i64 %indvars.iv397, %112, !dbg !21
%122 = getelementptr inbounds float, ptr %A, i64 %121, !dbg !21
%123 = load float, ptr %122, align 4, !dbg !21, !tbaa !37
%124 = fsub float %123, %110, !dbg !21
%125 = getelementptr inbounds float, ptr %T_subtract10, i64 %121, !dbg !21
store float %124, ptr %125, align 4, !dbg !21, !tbaa !63
%indvars.iv.next398 = add nuw nsw i64 %indvars.iv397, 1, !dbg !21
%exitcond401.not = icmp eq i64 %indvars.iv.next398, %wide.trip.count400,
. . . !dbg !21
br i1 %exitcond401.not, label
. . . %for_begin_ax352.for_end_ax354_crit_edge.us.us.us.us, label
. . . %for_body_ax353.us.us.us.us, !dbg !21, !prof !55, !llvm.loop !66

T F

middle.block612:

br i1 %cmp.n625, label
. . . %for_begin_ax352.for_end_ax354_crit_edge.us.us.us.us, label
. . . %for_body_ax353.us.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax352.for_end_ax354_crit_edge.us.us.us.us:

%indvars.iv.next403 = add nuw nsw i64 %indvars.iv402, 1, !dbg !21
%exitcond406.not = icmp eq i64 %indvars.iv.next403, %wide.trip.count405,
. . . !dbg !21
br i1 %exitcond406.not, label
. . . %for_begin_ax248.for_end_ax250_crit_edge.split.us.us.us.us, label
. . . %for_begin_ax352.preheader.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax248.for_end_ax250_crit_edge.split.us.us.us.us:

%indvars.iv.next408 = add nuw nsw i64 %indvars.iv407, 1, !dbg !21
%exitcond411.not = icmp eq i64 %indvars.iv.next408, %wide.trip.count410,
. . . !dbg !21
br i1 %exitcond411.not, label %for_begin_ax144.for_end_ax146_crit_edge.us,
. . . label %for_begin_ax248.preheader.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax160.for_end_ax162_crit_edge.us:

%indvars.iv.next433 = add nuw nsw i64 %indvars.iv432, 1, !dbg !21
%exitcond436.not = icmp eq i64 %indvars.iv.next433, %wide.trip.count435,
. . . !dbg !21
br i1 %exitcond436.not, label %for_begin_ax072.preheader, label
. . . %for_begin_ax160.preheader.us, !dbg !21, !prof !41

T F

for_begin_ax264.preheader.us.us.us:

%indvars.iv427 = phi i64 [%indvars.iv.next428,
. . . %for_begin_ax264.for_end_ax266_crit_edge.split.us.us.us.us], [0,
. . . %for_begin_ax160.preheader.us]
%131 = add nuw nsw i64 %indvars.iv427, %130
%132 = mul nuw nsw i64 %131, %129
br label %for_begin_ax368.preheader.us.us.us.us, !dbg !21

for_begin_ax368.preheader.us.us.us.us:

%indvars.iv422 = phi i64 [%indvars.iv.next423,
. . . %for_begin_ax368.for_end_ax370_crit_edge.us.us.us.us], [0,
. . . %for_begin_ax264.preheader.us.us.us]
%133 = add nuw nsw i64 %indvars.iv422, %132
%134 = mul nuw nsw i64 %133, %128
br i1 %min.iters.check629, label %for_body_ax369.us.us.us.us.preheader,
. . . label %vector.body633, !dbg !21, !prof !36

T F

for_body_ax369.us.us.us.us.preheader:

%indvars.iv417.ph = phi i64 [0, %for_begin_ax368.preheader.us.us.us.us],
. . . [%n.vec632, %middle.block627]
br label %for_body_ax369.us.us.us.us, !dbg !21

vector.body633:

%index634 = phi i64 [%index.next639, %vector.body633], [0,
. . . %for_begin_ax368.preheader.us.us.us.us], !dbg !21
%135 = add nuw nsw i64 %index634, %134, !dbg !21
%136 = getelementptr inbounds float, ptr %T_subtract10, i64 %135, !dbg !21
%137 = getelementptr inbounds nuw i8, ptr %136, i64 16, !dbg !21
%wide.load635 = load <4 x float>, ptr %136, align 4, !dbg !21, !tbaa !63
%wide.load636 = load <4 x float>, ptr %137, align 4, !dbg !21, !tbaa !63
%138 = getelementptr inbounds float, ptr %T_subtract5, i64 %135, !dbg !21
%139 = getelementptr inbounds nuw i8, ptr %138, i64 16, !dbg !21
%wide.load637 = load <4 x float>, ptr %138, align 4, !dbg !21, !tbaa !59
%wide.load638 = load <4 x float>, ptr %139, align 4, !dbg !21, !tbaa !59
%140 = fmul <4 x float> %wide.load635, %wide.load637, !dbg !21
%141 = fmul <4 x float> %wide.load636, %wide.load638, !dbg !21
store <4 x float> %140, ptr %138, align 4, !dbg !21, !tbaa !59
store <4 x float> %141, ptr %139, align 4, !dbg !21, !tbaa !59
%index.next639 = add nuw i64 %index634, 8, !dbg !21
%142 = icmp eq i64 %index.next639, %n.vec632, !dbg !21
br i1 %142, label %middle.block627, label %vector.body633, !dbg !21, !prof
. . . !52, !llvm.loop !67

T F

for_body_ax369.us.us.us.us:

%indvars.iv417 = phi i64 [%indvars.iv.next418,
. . . %for_body_ax369.us.us.us.us], [%indvars.iv417.ph,
. . . %for_body_ax369.us.us.us.us.preheader]
%143 = add nuw nsw i64 %indvars.iv417, %134, !dbg !21
%144 = getelementptr inbounds float, ptr %T_subtract10, i64 %143, !dbg !21
%145 = load float, ptr %144, align 4, !dbg !21, !tbaa !63
%146 = getelementptr inbounds float, ptr %T_subtract5, i64 %143, !dbg !21
%147 = load float, ptr %146, align 4, !dbg !21, !tbaa !59
%148 = fmul float %145, %147, !dbg !21
store float %148, ptr %146, align 4, !dbg !21, !tbaa !59
%indvars.iv.next418 = add nuw nsw i64 %indvars.iv417, 1, !dbg !21
%exitcond421.not = icmp eq i64 %indvars.iv.next418, %wide.trip.count420,
. . . !dbg !21
br i1 %exitcond421.not, label
. . . %for_begin_ax368.for_end_ax370_crit_edge.us.us.us.us, label
. . . %for_body_ax369.us.us.us.us, !dbg !21, !prof !55, !llvm.loop !68

T F

middle.block627:

br i1 %cmp.n640, label
. . . %for_begin_ax368.for_end_ax370_crit_edge.us.us.us.us, label
. . . %for_body_ax369.us.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax368.for_end_ax370_crit_edge.us.us.us.us:

%indvars.iv.next423 = add nuw nsw i64 %indvars.iv422, 1, !dbg !21
%exitcond426.not = icmp eq i64 %indvars.iv.next423, %wide.trip.count425,
. . . !dbg !21
br i1 %exitcond426.not, label
. . . %for_begin_ax264.for_end_ax266_crit_edge.split.us.us.us.us, label
. . . %for_begin_ax368.preheader.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax264.for_end_ax266_crit_edge.split.us.us.us.us:

%indvars.iv.next428 = add nuw nsw i64 %indvars.iv427, 1, !dbg !21
%exitcond431.not = icmp eq i64 %indvars.iv.next428, %wide.trip.count430,
. . . !dbg !21
br i1 %exitcond431.not, label %for_begin_ax160.for_end_ax162_crit_edge.us,
. . . label %for_begin_ax264.preheader.us.us.us, !dbg !21, !prof !41

T F

for_body_ax091.lr.ph:

%factor.op.mul280 = mul i32 %H, %W, !dbg !21
%.reass281 = mul i32 %factor.op.mul280, %N
%176 = sitofp i32 %.reass281 to float
%wide.trip.count460 = zext nneg i32 %C to i64, !dbg !21
%min.iters.check644 = icmp ult i32 %C, 4, !dbg !21
br i1 %min.iters.check644, label %for_body_ax091.preheader, label
. . . %vector.ph645, !dbg !21, !prof !36

T F

for_begin_k076.preheader.us.us.us.preheader:

%151 = zext nneg i32 %W to i64, !dbg !21
%152 = zext nneg i32 %H to i64, !dbg !21
%wide.trip.count455 = zext nneg i32 %C to i64, !dbg !21
%wide.trip.count450 = zext nneg i32 %N to i64
%wide.trip.count445 = zext nneg i32 %H to i64
%wide.trip.count440 = zext nneg i32 %W to i64
%xtraiter729 = and i64 %wide.trip.count440, 1
%153 = icmp eq i32 %W, 1
%unroll_iter733 = and i64 %wide.trip.count440, 2147483646
%lcmp.mod731.not = icmp eq i64 %xtraiter729, 0
br label %for_begin_k076.preheader.us.us.us, !dbg !21

for_begin_ax1114.preheader:

br i1 %18, label %for_body_ax1115.preheader, label
. . . %for_begin_ax0118.preheader, !dbg !21, !prof !29

T F

for_body_ax091.preheader:

%indvars.iv457.ph = phi i64 [0, %for_body_ax091.lr.ph], [%n.vec647,
. . . %middle.block642]
br label %for_body_ax091, !dbg !21

vector.ph645:

%n.vec647 = and i64 %wide.trip.count460, 2147483644, !dbg !21
%broadcast.splatinsert651 = insertelement <4 x float> poison, float %176,
. . . i64 0, !dbg !21
%broadcast.splat652 = shufflevector <4 x float> %broadcast.splatinsert651,
. . . <4 x float> poison, <4 x i32> zeroinitializer, !dbg !21
br label %vector.body648, !dbg !21

for_begin_k076.preheader.us.us.us:

%indvars.iv452 = phi i64 [0, %for_begin_k076.preheader.us.us.us.preheader
. . .], [%indvars.iv.next453,
. . . %for_begin_k076.for_end_k078_crit_edge.split.us.split.us.us.us.us]
%154 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv452
%.promoted.us.us.us276 = load float, ptr %154, align 4, !tbaa !32
br label %for_begin_k280.preheader.us.us.us.us.us, !dbg !21

for_begin_k280.preheader.us.us.us.us.us:

%indvars.iv447 = phi i64 [%indvars.iv.next448,
. . . %for_begin_k280.for_end_k282_crit_edge.split.us.us.us.us.us.us], [0,
. . . %for_begin_k076.preheader.us.us.us]
%.lcssa.us.lcssa.us.us264.us.us.us = phi float [%.lcssa,
. . . %for_begin_k280.for_end_k282_crit_edge.split.us.us.us.us.us.us], [
. . . %.promoted.us.us.us276, %for_begin_k076.preheader.us.us.us]
%155 = mul nuw nsw i64 %indvars.iv447, %0
%156 = add nuw nsw i64 %155, %indvars.iv452
%157 = mul nuw nsw i64 %156, %152
br label %for_begin_k384.preheader.us.us.us.us.us.us, !dbg !21

for_begin_k384.preheader.us.us.us.us.us.us:

%indvars.iv442 = phi i64 [%indvars.iv.next443,
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us], [0,
. . . %for_begin_k280.preheader.us.us.us.us.us]
%.lcssa.us257.us.us.us.us.us = phi float [%.lcssa,
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us], [
. . . %.lcssa.us.lcssa.us.us264.us.us.us, %for_begin_k280.preheader.us.us.us.us.us]
%158 = or i64 %indvars.iv442, %indvars.iv447
%159 = add nuw nsw i64 %indvars.iv442, %157
%160 = mul nuw nsw i64 %159, %151
%invariant.gep547 = getelementptr float, ptr %T_subtract5, i64 %160, !dbg !21
br i1 %153, label
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us.unr-lcssa, label
. . . %for_body_k385.us.us.us.us.us.us, !dbg !21, !prof !36

T F

for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us.unr-lcssa:

%.lcssa.ph = phi float [poison,
. . . %for_begin_k384.preheader.us.us.us.us.us.us], [%169,
. . . %for_body_k385.us.us.us.us.us.us]
%indvars.iv437.unr = phi i64 [0,
. . . %for_begin_k384.preheader.us.us.us.us.us.us], [%indvars.iv.next438.1,
. . . %for_body_k385.us.us.us.us.us.us]
%.unr730 = phi float [%.lcssa.us257.us.us.us.us.us,
. . . %for_begin_k384.preheader.us.us.us.us.us.us], [%169,
. . . %for_body_k385.us.us.us.us.us.us]
br i1 %lcmp.mod731.not, label
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us, label
. . . %for_body_k385.us.us.us.us.us.us.epil, !dbg !21, !prof !40

T F

for_body_k385.us.us.us.us.us.us:

%indvars.iv437 = phi i64 [%indvars.iv.next438.1,
. . . %for_body_k385.us.us.us.us.us.us], [0,
. . . %for_begin_k384.preheader.us.us.us.us.us.us]
%161 = phi float [%169, %for_body_k385.us.us.us.us.us.us], [
. . . %.lcssa.us257.us.us.us.us.us, %for_begin_k384.preheader.us.us.us.us.us.us]
%niter734 = phi i64 [%niter734.next.1, %for_body_k385.us.us.us.us.us.us],
. . . [0, %for_begin_k384.preheader.us.us.us.us.us.us]
%162 = or i64 %158, %indvars.iv437, !dbg !21
%163 = and i64 %162, 4294967295, !dbg !21
%164 = icmp eq i64 %163, 0, !dbg !21
%165 = select i1 %164, float 0.000000e+00, float %161, !dbg !21, !prof !26
%gep548 = getelementptr float, ptr %invariant.gep547, i64 %indvars.iv437,
. . . !dbg !21
%166 = load float, ptr %gep548, align 4, !dbg !21, !tbaa !59
%167 = fadd float %166, %165, !dbg !21
%indvars.iv.next438 = or disjoint i64 %indvars.iv437, 1, !dbg !21
%gep548.1 = getelementptr float, ptr %invariant.gep547, i64
. . . %indvars.iv.next438, !dbg !21
%168 = load float, ptr %gep548.1, align 4, !dbg !21, !tbaa !59
%169 = fadd float %168, %167, !dbg !21
%indvars.iv.next438.1 = add nuw nsw i64 %indvars.iv437, 2, !dbg !21
%niter734.next.1 = add i64 %niter734, 2, !dbg !21
%niter734.ncmp.1 = icmp eq i64 %niter734.next.1, %unroll_iter733, !dbg !21
br i1 %niter734.ncmp.1, label
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us.unr-lcssa, label
. . . %for_body_k385.us.us.us.us.us.us, !dbg !21, !prof !39

T F

for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us:

%.lcssa = phi float [%.lcssa.ph,
. . . %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us.unr-lcssa], [%175,
. . . %for_body_k385.us.us.us.us.us.us.epil], !dbg !21
%indvars.iv.next443 = add nuw nsw i64 %indvars.iv442, 1, !dbg !21
%exitcond446.not = icmp eq i64 %indvars.iv.next443, %wide.trip.count445,
. . . !dbg !21
br i1 %exitcond446.not, label
. . . %for_begin_k280.for_end_k282_crit_edge.split.us.us.us.us.us.us, label
. . . %for_begin_k384.preheader.us.us.us.us.us.us, !dbg !21, !prof !41

T F

for_body_k385.us.us.us.us.us.us.epil:

%170 = or i64 %158, %indvars.iv437.unr, !dbg !21
%171 = and i64 %170, 4294967295, !dbg !21
%172 = icmp eq i64 %171, 0, !dbg !21
%173 = select i1 %172, float 0.000000e+00, float %.unr730, !dbg !21, !prof
. . . !26

%gep548.epil = getelementptr float, ptr %invariant.gep547, i64
. . . %indvars.iv437.unr, !dbg !21
%174 = load float, ptr %gep548.epil, align 4, !dbg !21, !tbaa !59
%175 = fadd float %174, %173, !dbg !21
br label %for_begin_k384.for_end_k386_crit_edge.us.us.us.us.us.us, !dbg !21

for_begin_k280.for_end_k282_crit_edge.split.us.us.us.us.us.us:

%indvars.iv.next448 = add nuw nsw i64 %indvars.iv447, 1, !dbg !21
%exitcond451.not = icmp eq i64 %indvars.iv.next448, %wide.trip.count450,
. . . !dbg !21
br i1 %exitcond451.not, label
. . . %for_begin_k076.for_end_k078_crit_edge.split.us.split.us.us.us.us, label
. . . %for_begin_k280.preheader.us.us.us.us.us, !dbg !21, !prof !41

T F

for_begin_k076.for_end_k078_crit_edge.split.us.split.us.us.us.us:

store float %.lcssa, ptr %154, align 4, !tbaa !32
%indvars.iv.next453 = add nuw nsw i64 %indvars.iv452, 1, !dbg !21
%exitcond456.not = icmp eq i64 %indvars.iv.next453, %wide.trip.count455,
. . . !dbg !21
br i1 %exitcond456.not, label %for_body_ax091.lr.ph, label
. . . %for_begin_k076.preheader.us.us.us, !dbg !21, !prof !41

T F

for_body_ax091:

%indvars.iv457 = phi i64 [%indvars.iv.next458, %for_body_ax091], [
. . . %indvars.iv457.ph, %for_body_ax091.preheader]
%185 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv457,
. . . !dbg !21
%186 = load float, ptr %185, align 4, !dbg !21, !tbaa !32
%187 = fdiv float %186, %176, !dbg !21
store float %187, ptr %185, align 4, !dbg !21, !tbaa !32
%indvars.iv.next458 = add nuw nsw i64 %indvars.iv457, 1, !dbg !21
%exitcond461.not = icmp eq i64 %indvars.iv.next458, %wide.trip.count460,
. . . !dbg !21
br i1 %exitcond461.not, label %for_body_ax195.preheader, label
. . . %for_body_ax091, !dbg !21, !prof !55, !llvm.loop !72

T F

vector.body648:

%index649 = phi i64 [0, %vector.ph645], [%index.next653,
. . . %vector.body648], !dbg !21
%177 = getelementptr inbounds nuw float, ptr %A_red, i64 %index649, !dbg !21
%wide.load650 = load <4 x float>, ptr %177, align 16, !dbg !21, !tbaa !32
%178 = fdiv <4 x float> %wide.load650, %broadcast.splat652, !dbg !21
store <4 x float> %178, ptr %177, align 16, !dbg !21, !tbaa !32
%index.next653 = add nuw i64 %index649, 4, !dbg !21
%179 = icmp eq i64 %index.next653, %n.vec647, !dbg !21
br i1 %179, label %middle.block642, label %vector.body648, !dbg !21, !prof
. . . !42, !llvm.loop !70

T F

middle.block642:

%cmp.n654 = icmp eq i64 %n.vec647, %wide.trip.count460, !dbg !21
br i1 %cmp.n654, label %for_body_ax195.preheader, label
. . . %for_body_ax091.preheader, !dbg !21, !prof !46

T F

for_body_ax195.preheader:

%wide.trip.count465 = zext nneg i32 %C to i64, !dbg !21
%min.iters.check658 = icmp ult i32 %C, 8, !dbg !21
br i1 %min.iters.check658, label %for_body_ax195.preheader726, label
. . . %vector.ph659, !dbg !21, !prof !36

T F

for_body_ax195.preheader726:

%indvars.iv462.ph = phi i64 [0, %for_body_ax195.preheader], [%n.vec661,
. . . %middle.block656]
br label %for_body_ax195, !dbg !21

vector.ph659:

%n.vec661 = and i64 %wide.trip.count460, 2147483640, !dbg !21
br label %vector.body662, !dbg !21

for_body_ax195:

%indvars.iv462 = phi i64 [%indvars.iv.next463, %for_body_ax195], [
. . . %indvars.iv462.ph, %for_body_ax195.preheader726]
%191 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv462,
. . . !dbg !21
%192 = load float, ptr %191, align 4, !dbg !21, !tbaa !32
%193 = fadd float %192, 0x3EE4F8B580000000, !dbg !21
store float %193, ptr %191, align 4, !dbg !21, !tbaa !32
%indvars.iv.next463 = add nuw nsw i64 %indvars.iv462, 1, !dbg !21
%exitcond466.not = icmp eq i64 %indvars.iv.next463, %wide.trip.count465,
. . . !dbg !21
br i1 %exitcond466.not, label %for_body_i1.preheader, label %for_body_ax195,
. . . !dbg !21, !prof !55, !llvm.loop !75

T F

vector.body662:

%index663 = phi i64 [0, %vector.ph659], [%index.next666,
. . . %vector.body662], !dbg !21
%180 = getelementptr inbounds nuw float, ptr %A_red, i64 %index663, !dbg !21
%181 = getelementptr inbounds nuw i8, ptr %180, i64 16, !dbg !21
%wide.load664 = load <4 x float>, ptr %180, align 32, !dbg !21, !tbaa !32
%wide.load665 = load <4 x float>, ptr %181, align 16, !dbg !21, !tbaa !32
%182 = fadd <4 x float> %wide.load664, splat (float 0x3EE4F8B580000000),
. . . !dbg !21
%183 = fadd <4 x float> %wide.load665, splat (float 0x3EE4F8B580000000),
. . . !dbg !21
store <4 x float> %182, ptr %180, align 32, !dbg !21, !tbaa !32
store <4 x float> %183, ptr %181, align 16, !dbg !21, !tbaa !32
%index.next666 = add nuw i64 %index663, 8, !dbg !21
%184 = icmp eq i64 %index.next666, %n.vec661, !dbg !21
br i1 %184, label %middle.block656, label %vector.body662, !dbg !21, !prof
. . . !52, !llvm.loop !71

T F

middle.block656:

%cmp.n667 = icmp eq i64 %n.vec661, %wide.trip.count460, !dbg !21
br i1 %cmp.n667, label %for_body_i1.preheader, label
. . . %for_body_ax195.preheader726, !dbg !21, !prof !54

T F

for_body_i1.preheader:

%wide.trip.count470 = zext nneg i32 %C to i64, !dbg !21
%min.iters.check671 = icmp ult i32 %C, 4, !dbg !21
br i1 %min.iters.check671, label %for_body_i1.preheader725, label
. . . %vector.ph672, !dbg !21, !prof !36

T F

for_body_i1.preheader725:

%indvars.iv467.ph = phi i64 [0, %for_body_i1.preheader], [%n.vec674,
. . . %middle.block669]
br label %for_body_i1, !dbg !21

vector.ph672:

%n.vec674 = and i64 %wide.trip.count460, 2147483644, !dbg !21
br label %vector.body675, !dbg !21

for_body_i1:

%indvars.iv467 = phi i64 [%indvars.iv.next468, %for_body_i1], [
. . . %indvars.iv467.ph, %for_body_i1.preheader725]
%198 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv467,
. . . !dbg !21
%199 = load float, ptr %198, align 4, !dbg !21, !tbaa !32
%200 = tail call float @llvm.sqrt.f32(float %199), !dbg !21
store float %200, ptr %198, align 4, !dbg !21, !tbaa !32
%indvars.iv.next468 = add nuw nsw i64 %indvars.iv467, 1, !dbg !21
%exitcond471.not = icmp eq i64 %indvars.iv.next468, %wide.trip.count470,
. . . !dbg !21
br i1 %exitcond471.not, label %for_begin_ax098.preheader, label
. . . %for_body_i1, !dbg !21, !prof !55, !llvm.loop !76

T F

vector.body675:

%index676 = phi i64 [0, %vector.ph672], [%index.next678,
. . . %vector.body675], !dbg !21
%188 = getelementptr inbounds nuw float, ptr %A_red, i64 %index676, !dbg !21
%wide.load677 = load <4 x float>, ptr %188, align 16, !dbg !21, !tbaa !32
%189 = tail call <4 x float> @llvm.sqrt.v4f32(<4 x float> %wide.load677),
. . . !dbg !21
store <4 x float> %189, ptr %188, align 16, !dbg !21, !tbaa !32
%index.next678 = add nuw i64 %index676, 4, !dbg !21
%190 = icmp eq i64 %index.next678, %n.vec674, !dbg !21
br i1 %190, label %middle.block669, label %vector.body675, !dbg !21, !prof
. . . !42, !llvm.loop !74

T F

middle.block669:

%cmp.n679 = icmp eq i64 %n.vec674, %wide.trip.count460, !dbg !21
br i1 %cmp.n679, label %for_begin_ax098.preheader, label
. . . %for_body_i1.preheader725, !dbg !21, !prof !46

T F

for_body_ax1115.preheader:

%213 = zext nneg i32 %C to i64, !dbg !21
%214 = shl nuw nsw i64 %213, 2, !dbg !21
tail call void @llvm.memcpy.p0.p0.i64(ptr nonnull align 64 %T_subtract10,
. . . ptr align 64 %gamma, i64 %214, i1 false), !dbg !21
br label %for_begin_ax0118.preheader, !dbg !21

for_begin_ax0118.preheader:

br i1 %51, label %for_begin_ax1134.preheader, label
. . . %for_begin_ax1122.preheader.lr.ph, !dbg !21, !prof !36

T F

for_begin_ax2106.preheader.lr.ph:

%201 = mul nuw nsw i64 %indvars.iv487, %0
br i1 %brmerge568, label %for_end_ax1104, label
. . . %for_begin_ax2106.preheader.us.us, !prof !58

T F

for_end_ax1104:

%indvars.iv.next488 = add nuw nsw i64 %indvars.iv487, 1, !dbg !21
%exitcond491.not = icmp eq i64 %indvars.iv.next488, %wide.trip.count490,
. . . !dbg !21
br i1 %exitcond491.not, label %for_begin_ax1114.preheader, label
. . . %for_begin_ax1102.preheader, !dbg !21, !prof !41

T F

for_begin_ax2106.preheader.us.us:

%indvars.iv482 = phi i64 [%indvars.iv.next483,
. . . %for_begin_ax2106.for_end_ax2108_crit_edge.split.us.us.us], [0,
. . . %for_begin_ax2106.preheader.lr.ph]
%202 = add nuw nsw i64 %indvars.iv482, %201
%203 = mul nuw nsw i64 %202, %197
%204 = getelementptr inbounds nuw float, ptr %A_red, i64 %indvars.iv482
%205 = load float, ptr %204, align 4, !tbaa !32
%broadcast.splatinsert690 = insertelement <4 x float> poison, float %205,
. . . i64 0
%broadcast.splat691 = shufflevector <4 x float> %broadcast.splatinsert690,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax3110.preheader.us.us.us, !dbg !21

for_begin_ax3110.preheader.us.us.us:

%indvars.iv477 = phi i64 [%indvars.iv.next478,
. . . %for_begin_ax3110.for_end_ax3112_crit_edge.us.us.us], [0,
. . . %for_begin_ax2106.preheader.us.us]
%206 = add nuw nsw i64 %indvars.iv477, %203
%207 = mul nuw nsw i64 %206, %196
%invariant.gep549 = getelementptr float, ptr %T_subtract, i64 %207, !dbg !21
br i1 %min.iters.check683, label %for_body_ax3111.us.us.us.preheader, label
. . . %vector.body687, !dbg !21, !prof !36

T F

for_body_ax3111.us.us.us.preheader:

%indvars.iv472.ph = phi i64 [0, %for_begin_ax3110.preheader.us.us.us], [
. . . %n.vec686, %middle.block681]
br label %for_body_ax3111.us.us.us, !dbg !21

vector.body687:

%index688 = phi i64 [%index.next692, %vector.body687], [0,
. . . %for_begin_ax3110.preheader.us.us.us], !dbg !21
%208 = getelementptr float, ptr %invariant.gep549, i64 %index688, !dbg !21
%wide.load689 = load <4 x float>, ptr %208, align 4, !dbg !21, !tbaa !50
%209 = fdiv <4 x float> %wide.load689, %broadcast.splat691, !dbg !21
store <4 x float> %209, ptr %208, align 4, !dbg !21, !tbaa !50
%index.next692 = add nuw i64 %index688, 4, !dbg !21
%210 = icmp eq i64 %index.next692, %n.vec686, !dbg !21
br i1 %210, label %middle.block681, label %vector.body687, !dbg !21, !prof
. . . !42, !llvm.loop !77

T F

for_body_ax3111.us.us.us:

%indvars.iv472 = phi i64 [%indvars.iv.next473, %for_body_ax3111.us.us.us
. . .], [%indvars.iv472.ph, %for_body_ax3111.us.us.us.preheader]
%gep550 = getelementptr float, ptr %invariant.gep549, i64 %indvars.iv472,
. . . !dbg !21
%211 = load float, ptr %gep550, align 4, !dbg !21, !tbaa !50
%212 = fdiv float %211, %205, !dbg !21
store float %212, ptr %gep550, align 4, !dbg !21, !tbaa !50
%indvars.iv.next473 = add nuw nsw i64 %indvars.iv472, 1, !dbg !21
%exitcond476.not = icmp eq i64 %indvars.iv.next473, %wide.trip.count475,
. . . !dbg !21
br i1 %exitcond476.not, label
. . . %for_begin_ax3110.for_end_ax3112_crit_edge.us.us.us, label
. . . %for_body_ax3111.us.us.us, !dbg !21, !prof !55, !llvm.loop !78

T F

middle.block681:

br i1 %cmp.n693, label
. . . %for_begin_ax3110.for_end_ax3112_crit_edge.us.us.us, label
. . . %for_body_ax3111.us.us.us.preheader, !dbg !21, !prof !46

T F

for_begin_ax3110.for_end_ax3112_crit_edge.us.us.us:

%indvars.iv.next478 = add nuw nsw i64 %indvars.iv477, 1, !dbg !21
%exitcond481.not = icmp eq i64 %indvars.iv.next478, %wide.trip.count480,
. . . !dbg !21
br i1 %exitcond481.not, label
. . . %for_begin_ax2106.for_end_ax2108_crit_edge.split.us.us.us, label
. . . %for_begin_ax3110.preheader.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax2106.for_end_ax2108_crit_edge.split.us.us.us:

%indvars.iv.next483 = add nuw nsw i64 %indvars.iv482, 1, !dbg !21
%exitcond486.not = icmp eq i64 %indvars.iv.next483, %wide.trip.count485,
. . . !dbg !21
br i1 %exitcond486.not, label %for_end_ax1104, label
. . . %for_begin_ax2106.preheader.us.us, !dbg !21, !prof !41

T F

for_begin_ax1134.preheader:

br i1 %18, label %for_body_ax1135.preheader, label
. . . %for_begin_ax0138.preheader, !dbg !21, !prof !29

T F

for_begin_ax1122.preheader.lr.ph:

%215 = icmp slt i32 %H, 1
%216 = icmp slt i32 %W, 1
%217 = sext i32 %W to i64, !dbg !21
%218 = sext i32 %H to i64, !dbg !21
%wide.trip.count513 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count508 = zext nneg i32 %C to i64
%wide.trip.count503 = zext nneg i32 %H to i64
%wide.trip.count498 = zext i32 %W to i64
%brmerge571 = select i1 %215, i1 true, i1 %216
%min.iters.check697 = icmp ult i32 %W, 8
%n.vec700 = and i64 %wide.trip.count498, 2147483640
%cmp.n708 = icmp eq i64 %n.vec700, %wide.trip.count498
br label %for_begin_ax1122.preheader, !dbg !21

for_body_ax1135.preheader:

%233 = zext nneg i32 %C to i64, !dbg !21
%234 = shl nuw nsw i64 %233, 2, !dbg !21
tail call void @llvm.memcpy.p0.p0.i64(ptr nonnull align 64 %T_subtract5, ptr
. . . align 64 %beta, i64 %234, i1 false), !dbg !21
br label %for_begin_ax0138.preheader, !dbg !21

for_begin_ax0138.preheader:

br i1 %51, label %for_end_ax0140, label %for_begin_ax1142.preheader.lr.ph,
. . . !dbg !21, !prof !36

T F

for_begin_ax1122.preheader:

%indvars.iv510 = phi i64 [0, %for_begin_ax1122.preheader.lr.ph], [
. . . %indvars.iv.next511, %for_end_ax1124]
br i1 %18, label %for_begin_ax2126.preheader.lr.ph, label %for_end_ax1124,
. . . !dbg !21, !prof !29

T F

for_begin_ax2126.preheader.lr.ph:

%219 = mul nuw nsw i64 %indvars.iv510, %0
br i1 %brmerge571, label %for_end_ax1124, label
. . . %for_begin_ax2126.preheader.us.us, !prof !58

T F

for_end_ax1124:

%indvars.iv.next511 = add nuw nsw i64 %indvars.iv510, 1, !dbg !21
%exitcond514.not = icmp eq i64 %indvars.iv.next511, %wide.trip.count513,
. . . !dbg !21
br i1 %exitcond514.not, label %for_begin_ax1134.preheader, label
. . . %for_begin_ax1122.preheader, !dbg !21, !prof !41

T F

for_begin_ax2126.preheader.us.us:

%indvars.iv505 = phi i64 [%indvars.iv.next506,
. . . %for_begin_ax2126.for_end_ax2128_crit_edge.split.us.us.us], [0,
. . . %for_begin_ax2126.preheader.lr.ph]
%220 = getelementptr inbounds nuw float, ptr %T_subtract10, i64
. . . %indvars.iv505
%221 = add nuw nsw i64 %indvars.iv505, %219
%222 = mul nuw nsw i64 %221, %218
%223 = load float, ptr %220, align 4, !tbaa !63
%broadcast.splatinsert705 = insertelement <4 x float> poison, float %223,
. . . i64 0
%broadcast.splat706 = shufflevector <4 x float> %broadcast.splatinsert705,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax3130.preheader.us.us.us, !dbg !21

for_begin_ax3130.preheader.us.us.us:

%indvars.iv500 = phi i64 [%indvars.iv.next501,
. . . %for_begin_ax3130.for_end_ax3132_crit_edge.us.us.us], [0,
. . . %for_begin_ax2126.preheader.us.us]
%224 = add nuw nsw i64 %indvars.iv500, %222
%225 = mul nuw nsw i64 %224, %217
%invariant.gep551 = getelementptr float, ptr %T_subtract, i64 %225, !dbg !21
br i1 %min.iters.check697, label %for_body_ax3131.us.us.us.preheader, label
. . . %vector.body701, !dbg !21, !prof !36

T F

for_body_ax3131.us.us.us.preheader:

%indvars.iv495.ph = phi i64 [0, %for_begin_ax3130.preheader.us.us.us], [
. . . %n.vec700, %middle.block695]
br label %for_body_ax3131.us.us.us, !dbg !21

vector.body701:

%index702 = phi i64 [%index.next707, %vector.body701], [0,
. . . %for_begin_ax3130.preheader.us.us.us], !dbg !21
%226 = getelementptr float, ptr %invariant.gep551, i64 %index702, !dbg !21
%227 = getelementptr i8, ptr %226, i64 16, !dbg !21
%wide.load703 = load <4 x float>, ptr %226, align 4, !dbg !21, !tbaa !50
%wide.load704 = load <4 x float>, ptr %227, align 4, !dbg !21, !tbaa !50
%228 = fmul <4 x float> %broadcast.splat706, %wide.load703, !dbg !21
%229 = fmul <4 x float> %broadcast.splat706, %wide.load704, !dbg !21
store <4 x float> %228, ptr %226, align 4, !dbg !21, !tbaa !50
store <4 x float> %229, ptr %227, align 4, !dbg !21, !tbaa !50
%index.next707 = add nuw i64 %index702, 8, !dbg !21
%230 = icmp eq i64 %index.next707, %n.vec700, !dbg !21
br i1 %230, label %middle.block695, label %vector.body701, !dbg !21, !prof
. . . !52, !llvm.loop !79

T F

for_body_ax3131.us.us.us:

%indvars.iv495 = phi i64 [%indvars.iv.next496, %for_body_ax3131.us.us.us
. . .], [%indvars.iv495.ph, %for_body_ax3131.us.us.us.preheader]
%gep552 = getelementptr float, ptr %invariant.gep551, i64 %indvars.iv495,
. . . !dbg !21
%231 = load float, ptr %gep552, align 4, !dbg !21, !tbaa !50
%232 = fmul float %223, %231, !dbg !21
store float %232, ptr %gep552, align 4, !dbg !21, !tbaa !50
%indvars.iv.next496 = add nuw nsw i64 %indvars.iv495, 1, !dbg !21
%exitcond499.not = icmp eq i64 %indvars.iv.next496, %wide.trip.count498,
. . . !dbg !21
br i1 %exitcond499.not, label
. . . %for_begin_ax3130.for_end_ax3132_crit_edge.us.us.us, label
. . . %for_body_ax3131.us.us.us, !dbg !21, !prof !55, !llvm.loop !80

T F

middle.block695:

br i1 %cmp.n708, label
. . . %for_begin_ax3130.for_end_ax3132_crit_edge.us.us.us, label
. . . %for_body_ax3131.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax3130.for_end_ax3132_crit_edge.us.us.us:

%indvars.iv.next501 = add nuw nsw i64 %indvars.iv500, 1, !dbg !21
%exitcond504.not = icmp eq i64 %indvars.iv.next501, %wide.trip.count503,
. . . !dbg !21
br i1 %exitcond504.not, label
. . . %for_begin_ax2126.for_end_ax2128_crit_edge.split.us.us.us, label
. . . %for_begin_ax3130.preheader.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax2126.for_end_ax2128_crit_edge.split.us.us.us:

%indvars.iv.next506 = add nuw nsw i64 %indvars.iv505, 1, !dbg !21
%exitcond509.not = icmp eq i64 %indvars.iv.next506, %wide.trip.count508,
. . . !dbg !21
br i1 %exitcond509.not, label %for_end_ax1124, label
. . . %for_begin_ax2126.preheader.us.us, !dbg !21, !prof !41

T F

for_end_ax0140:

%259 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !21, !tbaa
. . . !22

%260 = tail call i32 %259(i32 1, i32 %dev_id, ptr nonnull %T_subtract10),
. . . !dbg !21
%.not = icmp eq i32 %260, 0, !dbg !21
br i1 %.not, label %if_end155, label %common.ret, !dbg !21, !prof !85

T F

for_begin_ax1142.preheader.lr.ph:

%235 = icmp slt i32 %H, 1
%236 = icmp slt i32 %W, 1
%237 = sext i32 %W to i64, !dbg !21
%238 = sext i32 %H to i64, !dbg !21
%wide.trip.count536 = zext nneg i32 %N to i64, !dbg !21
%wide.trip.count531 = zext nneg i32 %C to i64
%wide.trip.count526 = zext nneg i32 %H to i64
%wide.trip.count521 = zext i32 %W to i64
%brmerge574 = select i1 %235, i1 true, i1 %236
%min.iters.check712 = icmp ult i32 %W, 8
%n.vec715 = and i64 %wide.trip.count521, 2147483640
%cmp.n723 = icmp eq i64 %n.vec715, %wide.trip.count521
br label %for_begin_ax1142.preheader, !dbg !21

if_end155:

%261 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !21, !tbaa
. . . !22

%262 = tail call i32 %261(i32 1, i32 %dev_id, ptr nonnull %T_subtract5),
. . . !dbg !21
%.not166 = icmp eq i32 %262, 0, !dbg !21
br i1 %.not166, label %if_end158, label %common.ret, !dbg !21, !prof !85

T F

for_begin_ax1142.preheader:

%indvars.iv533 = phi i64 [0, %for_begin_ax1142.preheader.lr.ph], [
. . . %indvars.iv.next534, %for_end_ax1144]
br i1 %18, label %for_begin_ax2146.preheader.lr.ph, label %for_end_ax1144,
. . . !dbg !21, !prof !29

T F

for_begin_ax2146.preheader.lr.ph:

%239 = mul nuw nsw i64 %indvars.iv533, %0
br i1 %brmerge574, label %for_end_ax1144, label
. . . %for_begin_ax2146.preheader.us.us, !prof !58

T F

for_end_ax1144:

%indvars.iv.next534 = add nuw nsw i64 %indvars.iv533, 1, !dbg !21
%exitcond537.not = icmp eq i64 %indvars.iv.next534, %wide.trip.count536,
. . . !dbg !21
br i1 %exitcond537.not, label %for_end_ax0140, label
. . . %for_begin_ax1142.preheader, !dbg !21, !prof !41

T F

for_begin_ax2146.preheader.us.us:

%indvars.iv528 = phi i64 [%indvars.iv.next529,
. . . %for_begin_ax2146.for_end_ax2148_crit_edge.split.us.us.us], [0,
. . . %for_begin_ax2146.preheader.lr.ph]
%240 = getelementptr inbounds nuw float, ptr %T_subtract5, i64 %indvars.iv528
%241 = add nuw nsw i64 %indvars.iv528, %239
%242 = mul nuw nsw i64 %241, %238
%243 = load float, ptr %240, align 4, !tbaa !59
%broadcast.splatinsert720 = insertelement <4 x float> poison, float %243,
. . . i64 0
%broadcast.splat721 = shufflevector <4 x float> %broadcast.splatinsert720,
. . . <4 x float> poison, <4 x i32> zeroinitializer
br label %for_begin_ax3150.preheader.us.us.us, !dbg !21

for_begin_ax3150.preheader.us.us.us:

%indvars.iv523 = phi i64 [%indvars.iv.next524,
. . . %for_begin_ax3150.for_end_ax3152_crit_edge.us.us.us], [0,
. . . %for_begin_ax2146.preheader.us.us]
%244 = add nuw nsw i64 %indvars.iv523, %242
%245 = mul nuw nsw i64 %244, %237
br i1 %min.iters.check712, label %for_body_ax3151.us.us.us.preheader, label
. . . %vector.body716, !dbg !21, !prof !36

T F

for_body_ax3151.us.us.us.preheader:

%indvars.iv518.ph = phi i64 [0, %for_begin_ax3150.preheader.us.us.us], [
. . . %n.vec715, %middle.block710]
br label %for_body_ax3151.us.us.us, !dbg !21

vector.body716:

%index717 = phi i64 [%index.next722, %vector.body716], [0,
. . . %for_begin_ax3150.preheader.us.us.us], !dbg !21
%246 = add nuw nsw i64 %index717, %245, !dbg !21
%247 = getelementptr inbounds float, ptr %T_subtract, i64 %246, !dbg !21
%248 = getelementptr inbounds nuw i8, ptr %247, i64 16, !dbg !21
%wide.load718 = load <4 x float>, ptr %247, align 4, !dbg !21, !tbaa !50
%wide.load719 = load <4 x float>, ptr %248, align 4, !dbg !21, !tbaa !50
%249 = fadd <4 x float> %broadcast.splat721, %wide.load718, !dbg !21
%250 = fadd <4 x float> %broadcast.splat721, %wide.load719, !dbg !21
%251 = getelementptr inbounds float, ptr %T_add, i64 %246, !dbg !21
%252 = getelementptr inbounds nuw i8, ptr %251, i64 16, !dbg !21
store <4 x float> %249, ptr %251, align 4, !dbg !21, !tbaa !81
store <4 x float> %250, ptr %252, align 4, !dbg !21, !tbaa !81
%index.next722 = add nuw i64 %index717, 8, !dbg !21
%253 = icmp eq i64 %index.next722, %n.vec715, !dbg !21
br i1 %253, label %middle.block710, label %vector.body716, !dbg !21, !prof
. . . !52, !llvm.loop !83

T F

for_body_ax3151.us.us.us:

%indvars.iv518 = phi i64 [%indvars.iv.next519, %for_body_ax3151.us.us.us
. . .], [%indvars.iv518.ph, %for_body_ax3151.us.us.us.preheader]
%254 = add nuw nsw i64 %indvars.iv518, %245, !dbg !21
%255 = getelementptr inbounds float, ptr %T_subtract, i64 %254, !dbg !21
%256 = load float, ptr %255, align 4, !dbg !21, !tbaa !50
%257 = fadd float %243, %256, !dbg !21
%258 = getelementptr inbounds float, ptr %T_add, i64 %254, !dbg !21
store float %257, ptr %258, align 4, !dbg !21, !tbaa !81
%indvars.iv.next519 = add nuw nsw i64 %indvars.iv518, 1, !dbg !21
%exitcond522.not = icmp eq i64 %indvars.iv.next519, %wide.trip.count521,
. . . !dbg !21
br i1 %exitcond522.not, label
. . . %for_begin_ax3150.for_end_ax3152_crit_edge.us.us.us, label
. . . %for_body_ax3151.us.us.us, !dbg !21, !prof !55, !llvm.loop !84

T F

middle.block710:

br i1 %cmp.n723, label
. . . %for_begin_ax3150.for_end_ax3152_crit_edge.us.us.us, label
. . . %for_body_ax3151.us.us.us.preheader, !dbg !21, !prof !54

T F

for_begin_ax3150.for_end_ax3152_crit_edge.us.us.us:

%indvars.iv.next524 = add nuw nsw i64 %indvars.iv523, 1, !dbg !21
%exitcond527.not = icmp eq i64 %indvars.iv.next524, %wide.trip.count526,
. . . !dbg !21
br i1 %exitcond527.not, label
. . . %for_begin_ax2146.for_end_ax2148_crit_edge.split.us.us.us, label
. . . %for_begin_ax3150.preheader.us.us.us, !dbg !21, !prof !41

T F

for_begin_ax2146.for_end_ax2148_crit_edge.split.us.us.us:

%indvars.iv.next529 = add nuw nsw i64 %indvars.iv528, 1, !dbg !21
%exitcond532.not = icmp eq i64 %indvars.iv.next529, %wide.trip.count531,
. . . !dbg !21
br i1 %exitcond532.not, label %for_end_ax1144, label
. . . %for_begin_ax2146.preheader.us.us, !dbg !21, !prof !41

T F

if_end158:

%263 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !21, !tbaa
. . . !22

%264 = tail call i32 %263(i32 1, i32 %dev_id, ptr nonnull %T_subtract), !dbg
. . . !21

%.not167 = icmp eq i32 %264, 0, !dbg !21
br i1 %.not167, label %if_end161, label %common.ret, !dbg !21, !prof !85

T F

if_end161:

%265 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !21, !tbaa
. . . !22

%266 = tail call i32 %265(i32 1, i32 %dev_id, ptr nonnull %A_red), !dbg !21
%.not168 = icmp ne i32 %266, 0, !dbg !21
%. = sext i1 %.not168 to i32, !dbg !21
br label %common.ret, !dbg !21